



**THE UNIVERSITY
OF QUEENSLAND**
A U S T R A L I A

This exam paper must not be removed from the venue

Venue _____
 Seat Number _____
 Student Number

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 Family Name _____
 First Name _____

School of Civil Engineering
Semester Two Examinations, 2023
CIVL6415 Traffic Analysis and Simulation

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- This is a Closed Book examination - specified written materials permitted
- Casio FX82 series or UQ approved and labelled calculator only
- During Planning Time - Students are encouraged to review and plan responses to the exam questions
- This examination paper will be released to the Library

Materials Permitted in the Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

One A4 sheet of handwritten or typed notes double sided is permitted
 Unannotated Bilingual dictionary

Materials to be supplied to Students:

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x Multiple Choice Answer Sheet

Instructions to Students:

If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Please write your answers on the exam paper and the provided Multiple Choice Answer Sheet. Additional rough paper will be provided upon request.

For Examiner Use Only

Question Mark

Total _____

Part A: Multiple Choice Questions. To be completed in the multiple choice answer sheet. There is only one possible correct answer per question.

(Part A is worth 50/100 for the final exam. Each of the multiple choice questions is worth 2.5 marks.)

- 1) A traffic stream on a freeway has an average headway of 5s and a density of 30 veh/km. What is the space mean speed for this traffic stream?
- a) 20 km/h
 - b) 24 km/h
 - c) 28 km/h
 - d) 32 km/h
 - e) 36 km/h

- 2) A section of highway has the following flow-speed relationship, where density (k) is in veh/km and flow (q) is in veh/h.

$$q = 300v - 10v^2$$

What is the jam density?

- a) 100 veh/km
 - b) 150 veh/km
 - c) 200 veh/km
 - d) 250 veh/km
 - e) 300 veh/km
- 3) Which one of the statements below about the fundamental traffic flow variables is TRUE?
- a) Flow is the product of density and time mean speed.
 - b) Time mean speed might be equal to space mean speed under certain circumstances.
 - c) Time mean speed is the arithmetic mean of vehicle speeds at a given instant of time.
 - d) Flow is the inverse of average spacing.
 - e) The unit for occupancy is veh/km.
- 4) Three cars are travelling on a 5 km circular track at constant speeds of 90 km/h, 120 km/h and 150 km/h, respectively. What is the space-mean speed?
- a) 90 km/h
 - b) 105 km/h
 - c) 120 km/h
 - d) 135 km/h
 - e) 150 km/h
- 5) Which statement below about traffic models is TRUE?
- a) In static traffic models, travel time must be the same for all links.
 - b) In static traffic models, vehicles must be located only on one link.
 - c) Static traffic models do not allow alternative paths between origin and destination points.
 - d) In static traffic models, the relation between travel time and volume is nonlinear.
 - e) Static traffic models model individual vehicles in the network.

- 6) Which statement below about traffic models is TRUE?
- a) Mesoscopic models represent position of individual vehicles.
 - b) Mesoscopic models allow vehicles to have different speeds while travelling on the same link.
 - c) Macroscopic models are static models developed mainly for planning purposes.
 - d) Microscopic models build on continuous fluid approximation.
 - e) Microscopic models require an aggregated speed representation.
- 7) Which statement below about traffic simulation techniques is TRUE?
- a) Time-based simulation requires a list of upcoming events.
 - b) Event-based simulation requires pre-defined equal time intervals.
 - c) In time-based simulation, system state must be updated only when a new event occurs.
 - d) Event-based simulation models can cope with car following models.
 - e) Mesoscopic traffic models build on an event-based simulation technique.
- 8) Which statement below about transport systems is TRUE?
- a) Demand systems include infrastructure, vehicles and control technologies.
 - b) Modelling of long-term changes in transport systems may require reconsideration of land use (activity system) and major demand patterns.
 - c) The interaction between demand and supply is not considered in modelling of transport systems.
 - d) The level of congestion is the direct result of mode choices made by travellers.
 - e) None of the above.
- 9) Which statement below about IDM car following models is NOT true?
- a) IDM considers human factors through the desired speed and the desired space headway.
 - b) IDM is a very popular car following model because it is based on the safety-distance concept and always guarantees no collisions.
 - c) δ is a parameter in IDM and is usually set as 4.
 - d) When the preceding vehicle is far away, the distance-related term in the IDM equation becomes negligible and the model performs as a free flow model where the desired speed of the driver governs the acceleration.
 - e) When accelerating on a free road from a standstill, the vehicle described by IDM starts with the maximum acceleration.
- 10) Which statement below about car following is NOT true?
- a) Two primary goals of car following models are efficiency and sustainability of individual vehicles' movement.
 - b) The basic unit of car following models is the driver-vehicle.
 - c) Rules governing car following are typically simplified as a set of mathematical models that aim for reproducing the longitudinal interactions between a pair of vehicles.
 - d) Most of the existing car following models are continuous-time car following models.
 - e) Generally, discrete-time car following models are less accurate than continuous-time car following models; however, Gipps' car following model is discrete-time, and has been widely used.

- 11) Which statement below about GM car following models is NOT true?
- a) As one of the oldest car-following models, GM car following models are formulated from the human psychology perspective.
 - b) GM5 is a generic car following structure such that GM1-GM4 can be regarded as special cases.
 - c) According to GM4, two vehicles can get arbitrarily close as long as they are travelling at the same speed.
 - d) According to GM4, a vehicle at a stopped position will not be able to start moving again.
 - e) There exist inherent linkages between GM models and macroscopic traffic flow models.
- 12) Which statement below about Gipps' car following models is NOT true?
- a) Gipps' model is the most popular safety distance model by hypothesizing that the driver reacts to spacing relative to the preceding vehicle, rather than to the relative speed.
 - b) Gipps' model assumes that during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration).
 - c) Gipps' model assumes that there is a delay for the subject vehicle to execute acceleration or deceleration, and that during this delay, the subject vehicle cruises at a constant speed.
 - d) The safe speed estimated by Gipps' model can generally guarantee that a minimum gap between the subject vehicle and the preceding vehicle can be kept even if in the worse case scenario.
 - e) Gipps car following model is locally instable.
- 13) Traffic on a three-lane freeway observes a triangular flow-density relationship with free-flow speed $v_f = 100$ km/h, capacity $c = 1800$ veh/h/lane, jam density $k_j = 200$ veh/km/lane. Flow in the freeway from upstream is 3500 veh/h. At 9:00 a.m., an accident occurs on the freeway, the vehicles involved close one lane of traffic for 30 minutes and afterwards the scene is clear. At what time (h), does congestion disappear from the freeway?
- a) 9.25 h
 - b) 9.50 h
 - c) 9.75 h
 - d) Never; congestion does not disappear.
 - e) There is no congestion on the freeway.
- 14) Arrival traffic flow is 1200 veh/h, from time $t = 0$ to time $t = 1$ hr. After $t = 1$ hr, vehicles arrive at a (lesser) rate of 400 veh/h. Considering a capacity of 800 veh/h, what is the maximum queue length? Hint: Consider A-D curves.
- a) 200 veh
 - b) 300 veh
 - c) 400 veh
 - d) 500 veh
 - e) 600 veh
- 15) Which one of the statements below about macroscopic models is TRUE?
- a) First-order models account for driver anticipation.
 - b) First-order models consider the acceleration of traffic stream.
 - c) First-order models produce transient traffic states.
 - d) Second-order models account for both driver anticipation and reaction time.
 - e) Macroscopic models do not include any parameters to calibrate.

- 16) Which one of the statements below about ramp metering is TRUE?
- a) Ramp metering is implemented in signalised arterial road intersections.
 - b) Ramp metering is implemented via traffic signals on the off-ramps.
 - c) Ramp metering algorithms allow the vehicles to enter the freeway faster.
 - d) Ramp metering is useful, because it can prevent capacity drop.
 - e) Coordinated ramp metering algorithms are designed to make efficient use of storage capacity on one single on-ramp.
- 17) Which one of the statements below about equilibrium conditions is TRUE? Assume there is demand only within one O-D pair and there are multiple alternative paths.
- a) In stochastic user equilibrium conditions, only one path must be used.
 - b) In system optimum conditions, travel times on all the used routes must be equal.
 - c) In system optimum conditions, no user has an incentive to change their route choice.
 - d) In user equilibrium conditions, multiple paths can be used.
 - e) In stochastic user equilibrium conditions, all the used paths must have the same travel time.
- 18) Which one of the statements below about traffic assignment is TRUE?
- a) Dynamic traffic assignment problems require iterative methods that suffer significant convergence issues.
 - b) In dynamic traffic assignment context, the equilibrium problem cannot be formulated as a fixed-point solution.
 - c) Under user equilibrium conditions, every addition of capacity will improve traffic conditions.
 - d) Under system optimum conditions, increase in capacity may sometimes worsen traffic conditions.
 - e) None of the above.
- 19) Which one of the statements below about traffic assignment in Aimsun is TRUE?
- a) Aimsun does not allow enroute updates to vehicle routes.
 - b) Aimsun does not support iterative traffic assignment methods.
 - c) Aimsun is capable of considering experienced travel times when making route choices.
 - d) Aimsun offers a range of deterministic and stochastic route choice models.
 - e) None of the above.
- 20) Which one of the statements below about traffic assignment is TRUE?
- a) Traffic assignment concerns the selection of origins in the transportation network.
 - b) Deterministic models are the most advanced models that replicate the travel behaviour of commuters in urban areas.
 - c) Utility maximization rule states that an individual will select the alternative from the set of available alternatives that maximises their travel times.
 - d) In all-or-nothing assignment, trips from any origin zone to any destination zone are loaded into a single minimum cost path between them.
 - e) None of the above.

Part B: Short Answer Questions**Problem 1**

An observer standing at a point along a three-lane roadway collects spot speed data for all vehicles as they cross the observation point for 30 minutes. All vehicles in lane 1 are travelling at 50 km/h, all vehicles in lane 2 are travelling at 60 km/h, and all vehicles in lane 3 are travelling at 80 km/h. There is also a constant spacing of 50 m between vehicles. **[13 marks]**

- 1) What is the flow? **[4 marks]**
- 2) What is the time mean speed? **[3 marks]**
- 3) What is the space mean speed? **[3 marks]**
- 4) What is the density? **[3 marks]**

Problem 2

Consider two vehicles travelling on a road. Vehicle i is following vehicle $(i-1)$. The acceleration of vehicle i is described by the car-following model below:

$$a_i(t) = \frac{\beta[v_{i-1}(t) - v_i(t)]}{\alpha[x_{i-1}(t) - x_i(t)]}$$

Where $a_i(t)$ is the acceleration of vehicle i at time t , $v_i(t)$ and $v_{i-1}(t)$ are the speeds of vehicle i and $(i-1)$ at time t , $x_i(t)$ and $x_{i-1}(t)$ are the positions of vehicle i and $i-1$ at time t , and α and β are two parameters ($\alpha = 0.5$ s and $\beta = 0.3$ m.). Assuming the initial conditions are: the initial locations $x_{i-1}(t = 0) = 12$ m, $x_i(t = 0) = 2$ m and the initial speeds $v_{i-1}(t = 0) = 32$ m/s, $v_i(t = 0) = 36$ m/s. **[12 marks]**

- 1) In this car-following model, what are the stimuli, and how does each of these stimuli affect the estimated acceleration? **[4 marks]**
- 2) Determine the acceleration vehicle i will adopt at $t=0$ if this vehicle's behaviour can be perfectly described by this car-following model. **[4 marks]**
- 3) If you use this model to perform a one-step simulation at $t=0$ with a time step of 1 second, what will be the speed of vehicle i at $t=1$ second, and what will be the distance travelled by vehicle i from $t=0$ to $t=1$ second? Assume vehicle i travels at the constant acceleration within each time step. **[4 marks]**

Problem 3

Traffic on a three-lane freeway observes a triangular flow-density relationship with free-flow speed $v_f=100$ km/h, capacity $c=2000$ veh/h/lane, jam density $k_j=100$ veh/km/lane. Flow in the freeway from upstream is 5000 veh/h. At 9:00am, an accident occurs at point A. The vehicles involved close one lane of traffic (total capacity reduces to 4000 veh/h) for 30min and afterwards the scene is clear (capacity returns to 6000veh/h). **[15 marks]**

- 1) Sketch the space-time diagram with shockwaves (trajectories not needed). **[5 marks]**
- 2) Calculate the queue growth or dissipation rate in veh/h at 9:45am. **[5 marks]**
- 3) Mark enters the roadway via the ramp at point B (upstream of point A) at 9:30am. Distance AB is equal to 10 km. How many vehicles will be in the queue by the time Mark enters the congested traffic state? **[5 marks]**

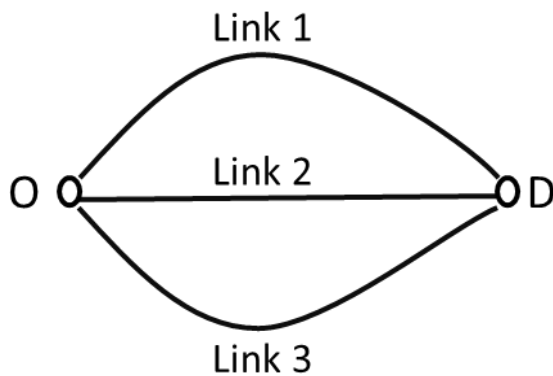
Problem 4

The figure below shows a network with 3 alternative links between an origin point O and a destination point D. The total demand between the two points is 10 veh, and the link performance functions (t_1 , t_2 , and t_3) are given below. Perform the first three iterations of Frank-Wolfe algorithm and complete the table below.

$t_{n,i}$ represents the travel time on link i at iteration n , $x_{n,i}$ represents the flow on link i at iteration n , and $y_{n,i}$ represents the auxiliary flow for link i at iteration n .

Hint: The formula for Frank-Wolfe update is $x_{n+1,i} = x_{n,i} + \alpha_n (y_{n,i} - x_{n,i})$

[10 marks]



$$t_1 = 10 \left(1 + 0.15 \left(\frac{x_1}{2} \right)^4 \right)$$

$$t_2 = 20 \left(1 + 0.15 \left(\frac{x_2}{4} \right)^4 \right)$$

$$t_3 = 25 \left(1 + 0.15 \left(\frac{x_3}{3} \right)^4 \right)$$

	Iterations - n			
	0	1	2	3
α_n	1.0	0.75	0.4	0.05
$t_{n,1}$	10			
$t_{n,2}$	20			
$t_{n,3}$	25			
$y_{n,1}$	10			
$y_{n,2}$	0			
$y_{n,3}$	0			
$x_{n,1}$	0			
$x_{n,2}$	0			
$x_{n,3}$	0			

END OF EXAMINATION